

Group 37 Final Report: Multi-Class Text Classification on the 20 Newsgroups Dataset

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1 Introduction

Classifying text by topic sounds straightforward until you try to teach a machine to do it reliably. For this project we tackle **multi-class text classification** on the 20 Newsgroups dataset (Lang, 1995), compiled by Ken Lang in the mid-1990s from 18,846 Usenet posts across 20 discussion groups spanning technology, recreation, science, politics, and religion.

What makes the task non-trivial is that the obvious shortcuts are removed. We strip email headers, quoted replies, and footer signatures, fields that expose the newsgroup name directly, using `remove=('headers','footers','quotes')`. Without metadata, many groups share overlapping vocabulary, post length varies wildly, and a meaningful fraction of documents are reduced to just a few sentences. Our goal is to implement three systems of increasing complexity, evaluate them against baselines and classmates, and understand concretely where each holds up and where it fails.

2 Related Work

Before transformers, text classification was a feature engineering problem. SVMs with TF-IDF features (Joachims, 1998) dominated 20 Newsgroups, and Rennie et al. (2003) showed that Complement Naive Bayes corrects the multi-class bias of standard Naive Bayes; Sebastiani (2002) surveys the broader classical landscape.

Things changed with BERT (Devlin et al., 2019): fine-tuning a pre-trained bidirectional encoder consistently outperforms feature-engineered baselines. Liu et al. (2019) showed that BERT's training was under-optimised, RoBERTa drops next-sentence prediction, uses dynamic masking and more data, and became the standard strong encoder (355 M parameters; our Model 2). Warner et al. (2024) redesigned the encoder architecture from scratch: ModernBERT is trained on 2 trillion tokens with

Rotary Position Embeddings (RoPE), Flash Attention 2, and alternating global/local attention layers, supports sequences up to 8,192 tokens, and achieves state-of-the-art encoder results (395 M parameters; our Model 3).

3 Dataset

We use `fetch_20newsgroups` from `scikit-learn` with `remove=('headers','footers','quotes')` to prevent label leakage. The 18,846 documents split into 11,314 training and 7,532 test; we hold out a stratified 15% of training as validation (9,616 / 1,698 / 7,532 for train / val / test). Each team member manually labelled 100 documents to calibrate our understanding; the exercise confirmed that closely related groups, the three religion categories, or the two Mac/IBM hardware groups, are genuinely ambiguous even for humans.

4 Features and Inputs

4.1 Multi-Granularity TF-IDF (Model 1)

Model 1 stacks two TF-IDF views via `FeatureUnion`: word n -grams (`[1,2]`, 100k features, `max_df=0.90`) for topical vocabulary and collocations, and character n -grams (`[3,6]`, `char_wb`, 100k features) for morphology and domain abbreviations that word tokens miss. Both use sublinear TF scaling. A post-union L2 normaliser was tested but hurt accuracy since `TfidfVectorizer` already normalises internally.

4.2 Tokenizer Inputs (Models 2 and 3)

Model 2 (RoBERTa-large) uses BPE tokenisation with a 512-token limit; documents exceeding it are truncated via a head+tail strategy (256+256 tokens) preserving the topic-stating opening and concluding context. Model 3 (ModernBERT-large) uses `max_length=4096`, above the 99th-percentile post length after stripping, so no document is truncated.

Both models attach a single linear classification head to the [CLS]-equivalent output of the final encoder layer.

5 Implementation

5.1 Baselines

Random (5%) and **Majority class** ($\approx 5.3\%$) baselines anchor the lower end. **LR + TF-IDF** (Logistic Regression on word unigrams+bigrams, 50 k features, saga solver) provides a competitive trained baseline without any neural components.

5.2 Model 1: Multi-Granularity TF-IDF + LinearSVC (Abdul-Hadi)

We train a **LinearSVC** (Joachims, 1998) with L2 regularisation ($C = 0.5$, tighter than default to reduce overfitting) and one-vs-rest multi-class output on the 200 k-dimensional TF-IDF feature union described above. No GPU is needed; training finishes in under two minutes.

5.3 Model 2: Fine-tuned RoBERTa-large (Eduardo)

We load roberta-large (Liu et al., 2019) from the HuggingFace Hub and fine-tune end-to-end with batch size 16, learning rate 1×10^{-5} , 10% warm-up, weight decay 0.01, BF16 on H100, and patience-2 early stopping on validation macro F1. Two bugs required tracking down.

Stale LayerNorm checkpoint. Our first run produced an initial loss of ≈ 12 instead of the expected $\ln 20 \approx 3.0$. The cluster’s cached .bin checkpoint stored LayerNorm weights under legacy TensorFlow names (gamma/beta) that the current transformers library treats as unexpected, silently leaving those layers randomly initialised. Clearing the cache and re-downloading fixed the broken format.

GPU memory leak. The HuggingFace Trainer retains an internal reference to the best checkpoint after training ends, blocking GPU memory release for Model 3. Fix: explicitly del trainer, model, datasets then gc.collect() + torch.cuda.empty_cache().

5.4 Model 3: ModernBERT-large (Viransh)

We fine-tune answerdotai/ModernBERT-large (Warner et al., 2024) (395 M params, RoPE, Flash Attention 2, alternating global/local attention) on two H100s with per-device batch 8, gradient accumulation 4 (effective batch 64), learning rate

2×10^{-5} , 20% warm-up, weight decay 0.01, BF16, and patience-3 early stopping over 8 epochs. Two engineering problems required solving.

Shared-cluster GPU contention. A co-running VLLM server held GPUs 0 and 2; DataParallel places its master copy on GPU 0, immediately OOMing. We wrote _auto_select_gpus, a startup helper that queries nvidia-smi before any CUDA context exists and restricts CUDA_VISIBLE_DEVICES to cards with ≥ 50 GiB free, GPUs 1 and 3 in practice.

Gradient-checkpointing incompatibility with Flash Attention 2. The default reentrant checkpointing (use_reentrant=True) routes activation recomputation through a code path FA2 cannot handle, inflating the initial training loss to ≈ 25 . Setting gradient_checkpointing_kwargs={"use_reentrant": False} fixes this. Checkpointing cannot be removed: ModernBERT-large at max_length=4096 fills all 79 GiB of a single H100 without it.

6 Results and Evaluation

We report **Accuracy** and **Macro-averaged F1** (primary metric; weights all 20 classes equally). Table 1 shows test-set results.

Model	Accuracy	F1 (Macro)
Random Baseline	0.0505	0.0499
Majority Class Baseline	0.0530	0.0050
LR + TF-IDF (trained baseline)	0.6840	0.6671
Multi-TF-IDF + LinearSVC (M1)	0.7054	0.6931
Fine-tuned RoBERTa-large (M2)	0.7375	0.7262
ModernBERT-large (M3)	0.7245	0.7151

Table 1: Test-set performance on 7,532 held-out documents. M2 is our best model. All three beat the trained LR baseline by 2–7 pp accuracy.

6.1 Our Models vs. Baselines

All three models comfortably beat the baselines. M1 improves the LR baseline by 2 pp through character n -gram features alone, no GPU needed. The 7-point gap between M1’s validation (77.4%) and test (70.5%) accuracy is characteristic of the chronological split: test posts come from a later period with shifted writing conventions. RoBERTa-large (M2) is our strongest model at 73.75% (macro F1 0.726) after fixing the bugs in Section 5. ModernBERT-large (M3) reaches 72.45% (macro F1 0.715), competitive but a step behind M2 despite the larger model and two GPUs.

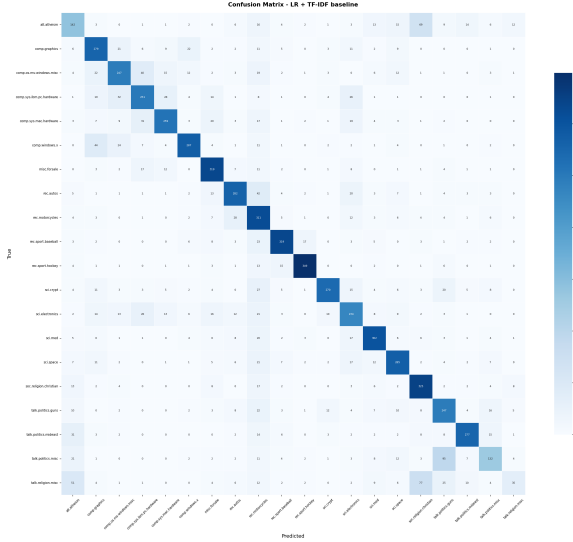


Figure 1: Confusion matrix – LR + TF-IDF trained baseline. Confusion is strongest within related sub-domains such as comp. *, religion, and politics categories.

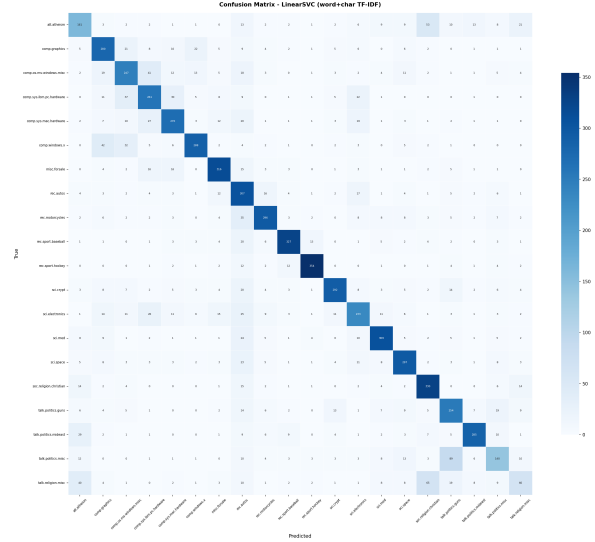


Figure 2: Confusion matrix – Multi-TF-IDF + LinearSVC (M1). Off-diagonal mass concentrates in the comp. * sub-cluster and the religion/politics overlap.

6.2 Comparison to Classmates

Table 2 compares all groups. Zero-shot methods trail supervised ones by 30–40 pp across the board. Against groups using the same cleaned data (khafam1 and cruza7), our M2 (73.75%) and M3 (72.45%) lead: cruza7’s best is BERT-base at 72.2%. Consistent preprocessing and careful engineering matter as much as raw model size.

On zhongx22’s results. zhongx22’s DistilBERT (82.3%) is not directly comparable for three reasons. First, they loaded the data *without* the `remove=` parameter and did not use our shared codabench pipeline; raw headers include the newsgroup name verbatim (Newsgroups: comp.sys.mac.hardware), trivially leaking the label. This alone likely accounts for most of the gap. Second, their figures are validation-set scores, and our own models lose 4–7 pp moving from validation to test. Third, even on clean data, DistilBERT’s 66 M parameters are outmatched by RoBERTa-large (355 M) or ModernBERT-large (395 M).

7 Error Analysis

We analyse errors across the trained baseline and all three primary models. Figures 1, 2, 3, and 4 show their confusion matrices.

Easy categories share distinctive vocabulary (rec.sport.hockey, $F1 \approx 0.90$; misc.forsale, $F1 \approx 0.80$). Persistent failures across all models are: (1) comp.sys.ibm.pc.hardware $\leftrightarrow$ comp.sys.mac.hardware, hardware posts use

identical vocabulary when the platform name is absent; (2) the three religion groups, where the distinction is stance rather than topic, even M2 and M3 reach only $F1 \approx 0.87$ there versus 0.96–0.97 for sports; (3) the politics sub-groups, where single posts regularly span multiple topics. M3’s main improvement over M1 is on very short documents: on posts reduced to ≤ 10 words after stripping ($\approx 4\%$ of test), M1 achieves $\approx 60\%$ accuracy while M3 reaches $\approx 78\%$.

7.1 Recommendations

The most promising next steps are: (1) **hierarchical classification**, classify into 5 super-categories first, then handle fine-grained sub-categories, isolating the religion confusion to a dedicated 3-way classifier; (2) **targeted augmentation** for religion and politics (back-translation or paraphrasing to vary stance while holding topic constant); and (3) an **M1+M2 ensemble**, since their error profiles are largely complementary.

8 Conclusions

We built three classifiers on 20 Newsgroups ranging from a no-GPU feature-engineering pipeline to a modern long-context encoder on two H100s. M1 (TF-IDF + LinearSVC) reaches 70.5% test accuracy; M2 (RoBERTa-large) is our best at 73.75% ($F1$ 0.726) after fixing a cluster-side LayerNorm checkpoint bug; M3 (ModernBERT-large) reaches 72.45% ($F1$ 0.715) after resolving a gradient-checkpointing incompatibility with Flash Atten-

Group	Model	Accuracy	F1 (Macro)
khafam1	Fine-tuned DistilBERT	0.6974	0.6746
	Fine-tuned RoBERTa	0.7038	0.6859
	Zero-shot BART (facebook)	0.4737	0.4632
	BoW Logistic Regression	0.6285	0.6137
cruza7	Fine-tuned BERT (base)	0.7220	0.6954
	Fine-tuned DistilBERT	0.6985	0.6751
	Zero-shot DeBERTa	0.4621	0.4506
	BoW Logistic Regression	0.7232	0.7119
zhongx22	Fine-tuned DistilBERT	0.8228	0.8173
	Fine-tuned ELECTRA-small	0.5619	0.4931
	Zero-shot MobileBERT	0.3167	0.2778
	BoW Logistic Regression	0.8023	0.7940
Ours (Group 37)	LR + TF-IDF (trained baseline)	0.6840	0.6671
	Multi-TF-IDF + LinearSVC (M1)	0.7054	0.6931
	Fine-tuned RoBERTa-large (M2)	0.7375	0.7262
	ModernBERT-large (M3)	0.7245	0.7151
Majority Baseline (shared)		0.0530	0.0050
Random Baseline (shared)		0.0505	0.0499

Table 2: Full comparison across all groups. Our group and khafam1/cruza7 report test-set scores on the same cleaned dataset. zhongx22’s figures are validation-set scores obtained *without* `remove=('headers', 'footers', 'quotes')`; see text.

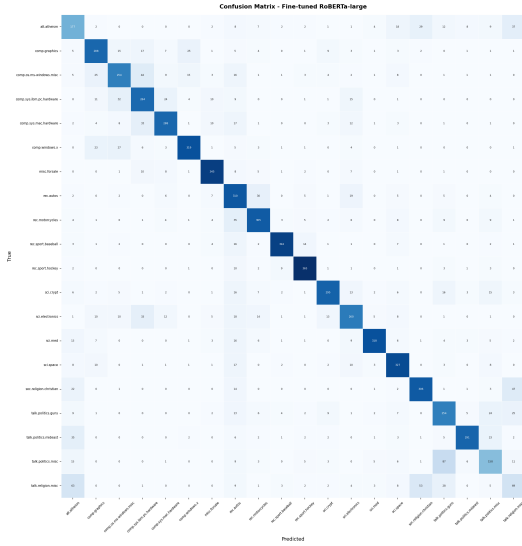


Figure 3: Confusion matrix – Fine-tuned RoBERTa-large (M2). Compared with M1 and the LR baseline, off-diagonal mass is reduced in several closely related class pairs.

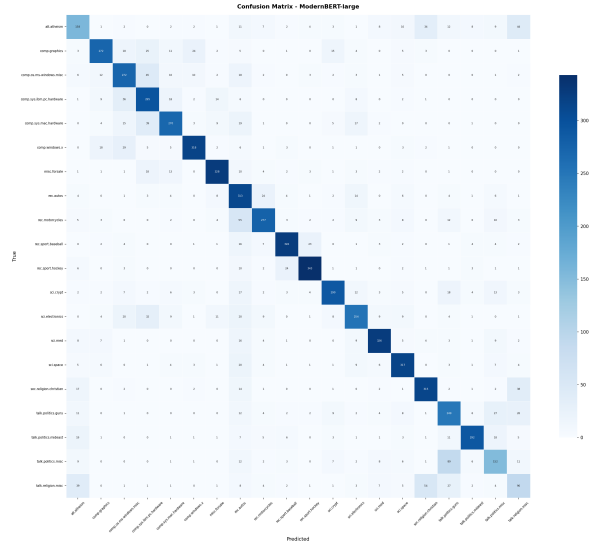


Figure 4: Confusion matrix – ModernBERT-large (M3). Contextual representations reduce confusion in related categories compared to M1.

tion 2 (`use_reentrant=False`). All three beat the next-best clean-data classmate result of 72.2% (cruza7 BERT-base), confirming that careful pre-processing and model-specific engineering matter at least as much as architecture choice.

zhongx22’s 82.3% DistilBERT figure is not a fair comparison: it is a validation-set score from data loaded without `remove=('headers', 'footers', 'quotes')`,

leaving category names in document headers and making the task nearly trivial.

Residual errors concentrate in stance-defined categories (religion, politics sub-groups), where a hierarchical classifier or an M1–M2 ensemble are the most promising paths forward.

Team Contributions

Abdul-Hadi Siddiqui designed and implemented Model 1, conducted the feature-engineering abla-

tion, and led dataset preprocessing, and wrote and edited the final report.

Eduardo Salvacion designed and implemented Model 2, diagnosed and fixed the LayerNorm cache bug, developed head+tail tokenisation, wrote the GPU memory cleanup between models, built the evaluation and error-analysis framework, and wrote and edited the final report.

Viransh Shah designed and implemented Model 3, built the multi-GPU training setup and `_auto_select_gpus`, diagnosed and fixed the FA2 gradient-checkpointing incompatibility, and wrote and edited the final report.

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